CURRICULUM VITAE
Helmut G. Katzgraber

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BIOGRAPHICAL DATA

Date and Place of Birth	July 7, 1972, Lima, Perú
Citizenship	American / Austrian / Peruvian
Languages	English / German / Spanish

EDUCATION

Ph.D. in Physics 06/1998 – 09/2001
University of California Santa Cruz (Advisor: Prof. A. Peter Young)

M.S. in Physics 10/1997 – 06/1998
University of California Santa Cruz (Advisor: Prof. A. Peter Young)

Diploma in Physics (M.S. equivalent) 10/1992 – 04/1997
ETH Zürich, Switzerland (Advisor: Prof. G. Blatter)
Awards: Diploma with distinction (*summa cum laude*), Pólya Prize

Military Service (Austria) 01/1992 – 09/1992
Österreichisches Bundesheer (Medal for outstanding service)

German Bachelors Degree 03/1989 – 12/1991
Colegio Alexander von Humboldt Lima, Perú (Award for best student in math & sciences)

RESEARCH & PROFESSIONAL EXPERIENCE

Venture Capital
55 North, Copenhagen, Denmark (Chief Science Officer & General Partner) 09/2025 –

Industry
AWS Professional Services, Seattle WA (Global Practice Lead – Quantum) 08/2020 – 08/2025
Microsoft, Redmond, WA (principal research manager, Quantum Solutions) 08/2018 – 08/2020

Board member
British Columbia Quantum Algorithms Institute (for Amazon), Vancouver, BC 01/2021 – 05/2022

Consultant
1QB Information Technologies (head of research, optimization), Vancouver, BC 02/2017 – 08/2018
Microsoft Quantum (consultant), Redmond, WA 06/2017 – 08/2018

Professor
Department of Physics & Astronomy, Texas A&M University, College Station, USA 09/2015 – 01/2021
Department of Physics, University of Washington, Seattle, WA, USA (affiliate position) 02/2019 – 08/2025

Faculty Member
External faculty member at the Santa Fe Institute, Santa Fe NM, USA 07/2014 – 07/2020
Visiting professor in computational physics, Coventry University, Coventry, UK 07/2015 – 07/2016
Materials Science and Engineering, Texas A&M University, College Station, USA 03/2013 – 03/2016

Associate Professor (with tenure)
Department of Physics & Astronomy, Texas A&M University, College Station, USA 09/2012 – 08/2015

Assistant Professor
Department of Physics & Astronomy, Texas A&M University, College Station, USA 01/2009 – 08/2012
Theoretical Physics Institute, ETH Zürich, Switzerland (SNF) 03/2007 – 03/2013

Post-Doctoral Researcher
Theoretical Physics Institute, ETH Zürich, Switzerland (Advisor: G. Blatter) 10/2002 – 03/2007
Department of Physics, University of California, Davis, USA (Advisor: G. Zimanyi) 10/2001 – 10/2002

Research Assistant
Department of Physics, University of California, Santa Cruz, USA 10/1997 – 10/2001
Theoretical Physics Institute, ETH Zürich, Switzerland 04/1997 – 09/1997

PUBLICATION & PRESENTATION SUMMARY

Publications (incl. patents, book chapters & conference proceedings)	205
Talks at conferences, seminars & colloquia	300+

EXECUTIVE EDUCATION

IMD Venture Capital Asset Management Program 05/2026 – 05/2026
Executive education to deepen skills in venture capital

Stanford University Graduate Business School LEAD Program 09/2020 – 09/2021
Executive education to deepen skills in economics, leadership, mentoring, and coaching

AWARDS

Fellow of the American Physical Society
Awarded by the DCOMP division of the APS 2021 (for contributions to computational physics methods)

Outstanding Referee Award
Awarded by the American Physical Society 2016

Distinguished Achievement College-Level Award in Teaching
Awarded by the Texas A&M Association of Former Students 2013

NSF CAREER Award
Awarded by the National Science Foundation 2012 (Faculty Early Career Development Program)

Teaching Excellence Award
Awarded by the Texas A&M System 2009

SNF Förderungsprofessur
Awarded by the Swiss National Science Foundation 2007 (faculty fellowship)

Pólya Prize
Awarded by ETH Zurich 1997 (best Diploma Thesis in mathematics and theoretical physics)

SCIENTIFIC VISITS & EXTERNAL APPOINTMENTS

Coventry University, Coventry, UK 07/2015 – 07/2016
International Visiting Professorship Scheme

Ben-Gurion University of the Negev, Beer Sheba, Israel 05/2014 – 06/2014
Distinguished Scientist Visitor Program

Chinese Academy of Sciences, Beijing, China 04/2012 – 05/2012
Scientific visit

Institute for Rock Magnetism, Minneapolis, MN, USA 05/2002
Experiments on disordered systems using a VSM

PUBLICATIONS (NOTE: STUDENTS UNDERLINED, POSTDOCS MARKED WITH A STAR)

Patents

24 Martin J. A. Schuetz, Ruben S. Andrist, and **Helmut G. Katzgraber**, "*Quantum-informed reduction algorithms for solving maximum independent set problems*"

23 Martin J. A. Schuetz, Ruben S. Andrist, Grant Salton, and **Helmut G. Katzgraber**, "*Quantum Compilation Toolkit For Combinatorial Optimization Problem Graphs*," (filed 03/2025)

22 Martin J. A. Schuetz, K. Brubaker, and **Helmut G. Katzgraber**, "*Decoder configured to enforce one or more constraints of a mixed integer program optimization problem*," (filed 12/2024)

21 Gili Rosenberg, J. K. Brubaker, Martin J. A. Schuetz, and **Helmut G. Katzgraber**, "*Machine Learning Model Pruning System,*" (filed 09/2024)

20 Thomas Haener and **Helmut G. Katzgraber**, "*Physics-based Process to Determine Configurations for Multi-Input, Multi-Output Boolean Functions,*" (filed (06/2023)

19 Damian S. Steiger, Thomas Haener and **Helmut G. Katzgraber**, "*Generating a Bounded Sparsity Ansatz Wavefunction For the Evaluation of a Ground State Energy of a Physical System*," (filed 03/2023)

18 G. Rosenberg, J. K. Brubaker, Zhihuai Zhu, Martin J. A. Schuetz, Thomas Haener, Grant Salton and **Helmut G. Katzgraber**, "*Solvers for Generating Interpretable Logical Boolean Formulas via Native Optimization*," (filed 03/2023)

17 Thomas Haener and **Helmut G. Katzgraber**, "*Physics-based process to determine configurations for multi-input monotone boolean functions*," (filed 12/2022)

16 Thomas Haener and **Helmut G. Katzgraber**, "*Sparse simulation of Clifford dominated quantum circuits*," (filed 12/2022)

15 Martin J. A. Schuetz, K. Brubaker, M. Resende, and **Helmut G. Katzgraber**, "*Random-key optimization for generating sequencing plans*," (filed 12/2022)

14 Martin J. A. Schuetz, G. Salton, and **Helmut G. Katzgraber**, "*Graph-based computational position assignments for solving optimization problems on quantum computing devices*," (filed 11/2022)

13 Jefferey C. Chancellor, Jared Taylor, and **Helmut G. Katzgraber**, "*System and Method for AI Based Spacecraft Shielding Design*," (filed 03/2022)

12 Martin J. A. Schuetz, J. Kyle Brubaker, and **Helmut G. Katzgraber**, "*Optimization for mixed-integer and continuous variable optimization problems with graph neural networks*" (filed 03/2022)

11 Michael Kastroyano, Nicola Pancotti, and **Helmut G. Katzgraber**, "*Tensor network solver for stochastic differential equations*," (filed 03/2022)

10 Martin J. A. Schuetz, J. Kyle Brubaker, and **Helmut G. Katzgraber**, "*Combinatorial optimization with graph neural networks*," (filed 06/2021)

9 Damian Steiger, Thomas Haener, Martin Roetteler and **Helmut G. Katzgraber**, "*Low-cost linear orders for quantum-program simulation,*" (filed 10/2020)

8 Firas Hamze and **Helmut G. Katzgraber**, "*Univariate density estimation method,*" (filed 10/2020)

7 Christopher Pattison, **Helmut G. Katzgraber** and Matthias Troyer, "*Cluster update accelerator circuit,*" (filed 01/2020)

6 Poulami Das, Nicolas Delfosse, Christopher Pattison, Srilatha Manne, Doug Carmean, Krysta Svore and **Helmut G. Katzgraber**, "*Pipelined Hardware Decoder for Quantum Computing Devices,*" (filed 12/2019)

5 Poulami Das, Nicolas Delfosse, Christopher Pattison, Srilatha Manne, Doug Carmean, Krysta Svore and **Helmut G. Katzgraber**, "*Systems for Coupling Decoders to Quantum Registers,*" (filed 12/2019)

4 Poulami Das, Nicolas Delfosse, Christopher Pattison, Srilatha Manne, Doug Carmean, Krysta Svore and **Helmut G. Katzgraber**, "*Syndrome Data Compression for Quantum Computing Devices,*" (filed 12/2019)

3 Damian S. Steiger, Christopher Pattison, **Helmut G. Katzgraber** and Matthias Troyer, "*Computing Stochastic Simulation Control Parameters,*" (filed 10/2019)

2 Maliheh Aramon Bajestani, Nicolas Pradignac, and **Helmut G. Katzgraber**, "*Method and system for solving mixed-integer programming problems using a feasibility pump technique embedded in a Monte Carlo simulation framework,*" (filed 06/2019)

1 Christopher Pattison, **Helmut G. Katzgraber** and Matthias Troyer, "*Accelerator for computing combinatorial cost function,*" (filed 02/2019)

Papers

167 Andrew J. Ochoa, Martin J. A. Schuetz, **Helmut G. Katzgraber**, "*Efficient Ground-State Manifold Amplification for Rydberg Atom Arrays via Houdayer Cluster Updates,*" (arXiv:2610.28528)

166 Romina Yalovetzky, Martin J. A. Schuetz, Zichang He, Jiayu Shen, Yue Sun, Rudy Raymond, Shauna Sahay, Kishore Perla, Ruben S. Andrist, Grant Salton, **Helmut G. Katzgraber**, Roger Bongiovanni, Niraj Kumar, Rob Otte, "*Quantum-Informed Portfolio Selection: An End-to-End Pipeline Validated on Trapped-Ion Hardware with Real Market Data,*" (arXiv:2607.01037)

165 A. Chaves, Mauricio G. C. Resende, Carise E. Schmidt, J. Kyle Brubaker, **Helmut G. Katzgraber**, Martin J. A. Schuetz, "*Applying a Random-Key Optimizer on Mixed Integer Programs*," (arXiv:2602.22173)

164 Serdar Kadioglu, Elton Yechao Zhu, Gili Rosenberg, John Kyle Brubaker, Martin J. A. Schuetz, Grant Salton, Zhihuai Zhu, **Helmut G. Katzgraber**, "*BoolXAI: Explainable AI Using Expressive Boolean Formulas*," Proceedings of the AAAI Conference on Artificial Intelligence 39(28), 28900 (2025)

163 Peter J. Eder, Aaron Kerschbaumer, Jernej Rudi Finzgar, Raimel A. Medina, Martin J. A. Schuetz, **Helmut G. Katzgraber**, Sarah Braun, Christian B. Mendl, "*Quantum-Guided Cluster Algorithms for Combinatorial Optimization*," (arXiv:2508.10656)

162 George Woodman, Ruben S. Andrist, Thomas Haener, Damian S. Steiger, Martin J. A. Schuetz, **Helmut G. Katzgraber**, Marcin Detyniecki, "*Modern Computational Methods in Reinsurance Optimization: From Simulated Annealing to Quantum Branch & Bound*," (arXiv:2504.16530)

161 Junpeng Hou, Amin Barzegar, **Helmut G. Katzgraber**, "*Direct comparison of stochastic driven nonlinear dynamical systems for combinatorial optimization,*" Phys. Rev. E 112, 035301 (2025)

160 Martin J. A. Schuetz, Romina Yalovetzky, Ruben S. Andrist, Grant Salton, Yue Sun, Rudy Raymond, Shouvanik Chakrabarti, Atithi Acharya, Ruslan Shaydulin, Marco Pistoia, **Helmut G. Katzgraber**, "*qReduMIS: A Quantum-Informed Reduction Algorithm for the Maximum Independent Set Problem,*" Phys. Rev. Research 8, 033296 (2026)

159 Martin J. A. Schuetz, Ruben S. Andrist, Grant Salton, Romina Yalovetzky, Rudy Raymond, Yue Sun, Atithi Acharya, Shouvanik Chakrabarti, Marco Pistoia, and **Helmut G. Katzgraber**, "*Compilation Toolkit for Rydberg Atom Arrays with Implications for Problem Hardness and Quantum Speedups*," Phys. Rev. Research 7, 033107 (2025)

158 J. Kyle Brubaker, Kyle E. Booth, Akihiko Arakawa, Fabian Furrer, Jayeeta Ghosh, Tsutomo Sato, **Helmut G. Katzgraber**, "*Quadratic unconstrained binary optimization and constraint programming approaches for lattice-based cyclic peptide docking*," Sci. Rep. 15, 20395 (2025)

157 Gili Rosenberg, J. Kyle Brubaker, Martin J. A. Schuetz, Elton Yechao Zhu, Serdar Kadioglu, Sima E. Borujeni, **Helmut G. Katzgraber**, "*Scalable iterative pruning of large language and vision models using block coordinate descent,*" (arXiv:2411.17796)

156 Antonio A. Chaves, Mauricio G. C. Resende, Martin J. A. Schuetz, J. Kyle Brubaker, **Helmut G. Katzgraber**, Edilson F. de Arruda, Ricardo M. A. Silva, "*A Random-Key Optimizer for Combinatorial Optimization*," (arXiv:2411.04293)

155 Atithi Acharya, Romina Yalovetzky, Pierre Minssen, Shouvanik Chakrabarti, Ruslan Shaydulin, Rudy Raymond, Yue Sun, Dylan Herman, Ruben S. Andrist, Grant Salton, Martin J. A. Schuetz, **Helmut G. Katzgraber**, and Marco Pistoia, *Decomposition Pipeline for Large-Scale Portfolio Optimization with Applications to Near-Term Quantum Computing*," Phys. Rev. Research 7, 023142 (2025)

154 Thomas Haener, Damian S. Steiger, Scott N. Genin, and **Helmut G. Katzgraber**, "*Sparse Simulation of VQE Circuits for Quantum Chemistry*," (arXiv:quant-phys/2404.10047)

153 Thomas Haener, Damian S. Steiger, and **Helmut G. Katzgraber**, "*Parallel Tempering for Logic Synthesis*," (arXiv:cs/2311.12394)

152 Jernej Rudi Finzgar, Aron Kerschbaumer, Martin J. A. Schuetz, Christian B. Mendl, and **Helmut G. Katzgraber**, "*Quantum-Informed Recursive Optimization Algorithms*," PRX Quantum 5, 020327 (2024)

151 Ruben S. Andrist, Martin J. A. Schuetz, Pierre Minssen, Romina Yalovetzky, Shouvanik Chakrabarti, Dylan Herman, Niraj Kumar, Grant Salton, Ruslan Shaydulin, Yue Sun, Marco Pistoia, **Helmut G. Katzgraber**, "*Hardness of the Maximum Independent Set Problem on Unit-Disk Graphs and Prospects for Quantum Speedups*," Phys. Rev. Research 5, 043277 (2023)

150 Gili Rosenberg, J. Kyle Brubaker, Martin J. A. Schuetz, Grant Salton, Zhihuai Zhu, Elton Yechao Zhu, Serdar Kadıoglu, Sima E. Borujeni, **Helmut G. Katzgraber**, "*Explainable AI using expressive Boolean formulas*," Mach. Learn. Knowl. Extr. 2023, 5(4), 1760-1795

149 Jernej Rudi Finzgar, Martin J. A. Schuetz, J. Kyle Brubaker, Hidetoshi Nishimori, and **Helmut G. Katzgraber**, "*Designing Quantum Annealing Schedules using Bayesian Optimization*," Phys. Rev. Research 6, 023063 (2024)

148 Martin J. A. Schuetz, J. Kyle Brubaker and **Helmut G. Katzgraber**, "*Reply to: Inability of a graph neural network heuristic to outperform greedy algorithms in solving combinatorial optimization problems*," by Stefan Boettcher Nat. Mach. Intell. (2022)

147 Martin J. A. Schuetz, J. Kyle Brubaker and **Helmut G. Katzgraber**, "*Reply to: Modern graph neural networks do worse than classical greedy algorithms in solving combinatorial optimization problems like maximum independent set*," by Maria Chiara Angelini and Federico Ricci-Tersenghi Nat. Mach. Intell. (2022)

146 Alexander M. Dalzell, B. David Clader, Grant Salton, Mario Berta, Cedric Yen-Yu Lin, David A. Bader, Nikitas Stamatopoulos, Martin J. A. Schuetz, Fernando G. S. L. Brandao, **Helmut G. Katzgraber**, William J. Zeng, "*End-to-end resource analysis for quantum interior point methods and portfolio optimization*," PRX Quantum 4, 040325 (2023)

145 Martin J. A. Schuetz, J. Kyle Brubaker, Henry Montagu, Yannick van Dijk, Johannes Klepsch, Philipp Ross, Andre Luckow, Mauricio G. C. Resende and **Helmut G. Katzgraber**, "*Optimization of Robot Trajectory Planning with Nature-Inspired and Hybrid Quantum Algorithms*," Phys. Rev. Applied 18, 054045 (2022)

144 Martin J. A. Schuetz, J. Kyle Brubaker, Zhihuai Zhu, and **Helmut G. Katzgraber**, "*Graph Coloring with Physics-Inspired Graph Neural Networks*," Phys. Rev. Research 4, 043131 (2022)

143 Martin J. A. Schuetz, J. Kyle Brubaker and **Helmut G. Katzgraber**, "*Combinatorial Optimization with Physics-Inspired Graph Neural Networks*," Nat. Mach. Intell. 4, 367 (2022)

142 Stephen P. Jordan, Siyuan Hu, Ignacio Rozada, Debra F. McGivney, Rasim Boyacioglu, Darryl C. Jacob, Sherry Huang, Michael Beverland, **Helmut G. Katzgraber**, Matthias Troyer, Mark A. Griswold, Dan Ma, "*Automated design of pulse sequences for magnetic resonance fingerprinting using physics-inspired optimization*," Proc. Nat. Acad. Sci. 118, e2020516118 (2021)

141 Mario S. Koenz, Wolfgang Lechner, **Helmut G. Katzgraber**, Matthias Troyer, "*Scaling overhead of embedding optimization problems in quantum annealing,*" Phys. Rev. X Quantum 2, 040322 (2021)

140 Amin Barzegar,* Anuj Kankani, Salvatore Mandrà, **Helmut G. Katzgraber**, "*Optimization and benchmarking of the thermal cycling algorithm,*" Phys. Rev. E 104, 035302 (2021)

139 Elisabetta Valiante, Maritza Hernandez, Amin Barzegar,* **Helmut G. Katzgraber**, "*Scaling overhead of locality reduction in binary optimization problems,*" Comp. Phys. Comm. 269, 108102 (2021)

138 Kohji Nishimura, Hidetoshi Nishimori, **Helmut G. Katzgraber**, "*Griffiths-McCoy singularity on the diluted Chimera graph: Monte Carlo simulations and experiments on the quantum hardware,*" Phys. Rev. A 102, 042403 (2020)

137 Dilina Perera,* Inimfon Akpabio, Firas Hamze, Salvatore Mandra, Nathan Rose, Maliheh Aramon, **Helmut G. Katzgraber**, "*Chook – A comprehensive suite for generating binary optimization problems with planted solutions,*" (arxiv:quant-ph/2005.14344)

136 Katja Biswas* and **Helmut G. Katzgraber**, "*Adding color: Visualization of energy landscapes in spin glasses,*" (arxiv:cond-mat/2004.12431)

135 Chao Fang, Amin Barzegar, and **Helmut G. Katzgraber**, "*Machine learning in physics: The pitfalls of poisoned training sets,*" Mach. Learn.: Sci. Technol. 1 045001

134 Dilina Perera,* Firas Hamze, Jack Raymond, Martin Weigel, and **Helmut G. Katzgraber**, "*Computational hardness of spin-glass problems with tile-planted solutions,*" Phys. Rev. E 101, 023316 (2020)

133 Ignacio Rozada, Maliheh Aramon, Jonathan Machta and **Helmut G. Katzgraber**, "*Effects of setting the temperatures in the parallel tempering Monte Carlo algorithm,*" Phys. Rev. E 100, 043311 (2019)

132 Nicolas Pradignac, Maliheh Aramon and **Helmut G. Katzgraber**, "*A feasibility pump algorithm embedded in an annealing framework,*" (arXiv:math.OC/1906.06434)

131 Firas Hamze, Jack Raymond, Christopher A. Pattison, Katja Biswas,* **Helmut G. Katzgraber**, "*The Wishart planted ensemble: A tunably-rugged pairwise Ising model with a first-order phase transition,*" Phys. Rev. E 101, 052102 (2020)

130 Zheng Zhu and **Helmut G. Katzgraber**, "*Do tensor renormalization group methods work for frustrated spin systems?,*" (arxiv:cond-mat/1903.07721)

129 Zheng Zhu, Andrew J. Ochoa, **Helmut G. Katzgraber**, "*Fair sampling of ground-state configurations of binary optimization problems,*" Phys. Rev. E 99, 063314 (2019)

128 Humberto Munoz-Bauza, Firas Hamze, **Helmut G. Katzgraber**, "*Learning to find order in disorder,*" J. Stat. Mech. 073302 (2020)

127 Philipp Hauke, **Helmut G. Katzgraber**, Wolfgang Lechner, Hidetoshi Nishimori, William D. Oliver, "*Perspectives of quantum annealing: Methods and implementations,*" Rep. Prog. Phys. 83 054401 (2020)

126 Amin Barzegar, Juan Carlos Andresen, Moshe Schechter, and **Helmut G. Katzgraber**, "*Numerical observation of a glassy phase in the three-dimensional Coulomb glass,*" Phys. Rev. B 100, 104418 (2019)

125 John Ferre, Amin Barzegar, **Helmut G. Katzgraber**, and Richard Scalettar, "*Distribution of inter-event avalanche times in disordered and frustrated spin systems,*" Phys. Rev. B 99, 024411 (2019)

124 Maliheh Aramon, Gili Rosenberg, Elisabetta Valiante, Toshiyuki Miyazawa, Hirotaka Tamura, and **Helmut G. Katzgraber**, "*Physics-Inspired Optimization for Quadratic Unconstrained Problems Using a Digital Annealer,*" Front. Phys. 7, 48 (2019)

123 Mario S. Koenz, Guglielmo Mazzola, Andrew J. Ochoa, **Helmut G. Katzgraber** and Matthias Troyer, "*Uncertain Fate of Fair Sampling in Quantum Annealing,*" Phys. Rev. A 100, 030303(R) (2019)

122 **Helmut G. Katzgraber** and M. A. Novotny, "*A small-world search for quantum speedup: How small-world interactions can lead to improved quantum annealer designs,*" Phys. Rev. Applied 10, 054004 (2018)

121 Andrew J. Ochoa, Darryl C. Jacob, Salvatore Mandra, and **Helmut G. Katzgraber**, "*Feeding the Multitude: A Polynomial-time Algorithm to Improve Sampling,*" Phys. Rev. E 99, 043306 (2019)

120 Chao Fang, Zheng Zhu, and **Helmut G. Katzgraber**, "*NAE-SAT-based probabilistic membership filters,*" arXiv:cs.DS/1801.06232

119 Wenlong Wang,* M. A. Moore, and **Helmut G. Katzgraber**, "*Fractal dimension of interfaces in Edwards-Anderson spin glasses for up to six space dimensions,*" Phys. Rev. E 97, 032104 (2018)

118 Firas Hamze, Darryl C. Jacob, Andrew J. Ochoa, Dilina Perera,* Wenlong Wang,* and **Helmut G. Katzgraber**, "*From Near to Eternity: Spin-glass planting, tiling puzzles, and constraint satisfaction problems,*" Phys. Rev. E 97, 043303 (2018)

117 S. Mandrà and **Helmut G. Katzgraber**, "*A deceptive step towards quantum speedup detection,*" Quant. Sci. Technol. 3, 04LT01 (2018)

116 Amin Barzegar, Christopher Pattison, Wenlong Wang,* and **Helmut G. Katzgraber**, "*Optimization of population annealing Monte Carlo for large-scale spin-glass simulations,*" Phys. Rev. E 98, 053308 (2018)

115 Wenlong Wang,* Jonathan Machta, Humberto Munoz-Bauza, **Helmut G. Katzgraber**, "*Number of thermodynamic states in the three-dimensional Edwards-Anderson spin glass,*" Phys. Rev. B 96, 184417 (2017)

114 Jeffery C. Chancellor, Rebecca S. Blue, Keith A. Cengel, Serena M. Aunon, Kathleen H. Rubins, **Helmut G. Katzgraber**, and Ann R. Kennedy, "*Limitations in Predicting the Space Radiation Health Risk for Exploration Astronauts,*" Nature Microgravity 4, 8 (2018)

113 Alejandro Perdomo-Ortiz, Alexander Feldman, Asier Ozaeta, Sergei V. Isakov, Zheng Zhu,* Bryan O'Gorman, **Helmut G. Katzgraber**, Alexander Diedrich, Hartmut Neven, Johan de Kleer, Brad Lackey, Rupak Biswas, "*On the readiness of quantum optimization machines for industrial applications,*" Phys. Rev. Applied 12, 014004 (2019) ▶ Editor's Suggestion

112 **Helmut G. Katzgraber**, "*Viewing Vanilla Quantum Annealing Through Spin Glasses,*" Quantum Sci. Technol. 3, 030505 (2018) ▶ invited perspective article on "*What would you do with 1000 qubits?* (most read article of 2018)"

111 Hamed Karimi, Gili Rosenberg, **Helmut G. Katzgraber**, "*Effective optimization using sample persistence: A case study on quantum annealers and various Monte Carlo optimization methods,*" Phys. Rev. E 96, 043312 (2017) ▶ graphics selected for the Phys. Rev. E Kaleidoscope (Nov. 2017)

110 Juan Carlos Andresen, **Helmut G. Katzgraber**, and Moshe Schechter, "*Random-field-induced disordering mechanism in a disordered ferromagnet: Between the Imry-Ma and the standard disordering mechanism,*" Phys. Rev. B 96, 214414 (2017)

109 Jeffery C. Chancellor, Stephen Guetersloh, Keith Cengel, John Ford, and **Helmut G. Katzgraber**, "*Emulation of the space radiation environment for materials testing and radiobiological experiments,*" Phys. Rev. Appl., submitted (arXiv:app-phys/1706.02727)

108 Wenlong Wang,* Salvatore Mandrà, and **Helmut G. Katzgraber**, "*Patch-planting spin-glass solution for benchmarking,*" Phys. Rev. E 96, 023312 (2017)

107 Wenlong Wang,* M. A. Moore, and **Helmut G. Katzgraber**, "*The Fractal Dimension of Interfaces in Edwards-Anderson and Long-range Ising Spin Glasses: Determining the Applicability of Different Theoretical Descriptions,*" Phys. Rev. Lett. 119, 100602 (2017)

106 S. Mandrà, **Helmut G. Katzgraber** and Creighton Thomas, "*The pitfalls of planar spin-glass benchmarks: Raising the bar for quantum annealers (again),*" Quantum Sci. Technol. 2, 038501, (2017) ► selected as one of 10 articles to be featured in a highlights collection for the journal's first year of publication

105 R. Santana, Z. Zhu,* and **Helmut G. Katzgraber**, "*Evolutionary Approaches to Optimization Problems in Chimera Topologies,*" enetic and Evolutionary Computation Conference (GECCO-2016), ACM Press, p. 397-404 (2016)

104 S. Mandrà, Z. Zhu,* and **Helmut G. Katzgraber**, "*Exponentially-Biased Ground-State Sampling of Quantum Annealing Machines with Transverse-Field Driving Hamiltonians,*" Phys. Rev. Lett. 118, 070502 (2017)

103 T. Aspelmeier, Wenlong Wang,* M. A. Moore, and **Helmut G. Katzgraber**, "*The interface free-energy exponent in the one-dimensional Ising spin glass with long-range interactions in both the droplet and broken replica symmetry regions,*" Phys. Rev. E, 94, 022116 (2016)

102 Z. Zhu,* Chao Fang and **Helmut G. Katzgraber**, "*borealis - A generalized global update algorithm for Boolean optimization problems,*" arXiv:physics.comp-ph/1605.09399 ► winner of the incomplete unweighted Max-SAT random track of the 2016 SAT competition (July 2016)

101 Kohji Nishimura, Hidetoshi Nishimori, Andrew J. Ochoa and **Helmut G. Katzgraber**, "*Retrieving the ground state of spin glasses using thermal noise: Performance of quantum annealing at finite temperatures,*" Phys. Rev. E 94, 032105 (2016)

100 Z. Zhu,* Andrew J. Ochoa and **Helmut G. Katzgraber**, "*Lack of a thermodynamic finite-temperature spin-glass phase in the two-dimensional randomly-coupled ferromagnet,*" Phys. Rev. B 97, 174425 (2018)

99 S. Mandrà, Z. Zhu,* W. Wang,* A. Perdomo-Ortiz, **Helmut G. Katzgraber**, "*Strengths and Weaknesses of Weak-Strong Cluster Problems: A Detailed Overview of State-of-the-art Classical Heuristics vs Quantum Approaches,*" Phys. Rev. A 94, 022337 (2016)

98 Ruben S. Andrist,* **Helmut G. Katzgraber**, H. Bombin, M. A. Martin-Delgado, "*Error tolerance of topological codes with independent bit-flip and measurement errors,*" Phys. Rev. A 94, 012318 (2016)

97 Daniel M. Packwood, Helmut G. Katzgraber, and Winfried Teizer, "*Stochastic Boltzmann Equation for Magnetic Relaxation in High-Spin Molecules,*" Proc. R. Soc. A 472 20150699 (2016)

96 Wenlong Wang,* Jonathan Machta, **Helmut G. Katzgraber**, "*Bond chaos in spin glasses revealed through thermal boundary conditions,*" Phys. Rev. B 93, 224414 (2016)

95 O. Melchert,* **Helmut G. Katzgraber**, M. A. Novotny, "*Site and bond percolation thresholds in $K_{n,n}$-based lattices: Vulnerability of quantum annealers to random qubit and coupler failures on Chimera topologies,*" Phys. Rev. E 93, 042128 (2016)

94 T. Aspelmeier, **Helmut G. Katzgraber**, Derek Larson, M. A. Moore, M. Wittmann, Joonhyun Yeo, "*Finite-size critical scaling in Ising spin glasses in the mean-field regime,*" Phys. Rev. E 93, 032123 (2016)

93 Wenlong Wang,* Jonathan Machta, **Helmut G. Katzgraber**, "*Population Annealing: Theory and Application in Spin Glasses,*" Phys. Rev. E 92, 063307 (2015)

92 Wenlong Wang, Jonathan Machta, **Helmut G. Katzgraber**, "*Chaos in Spin Glasses Revealed Through Thermal Boundary Conditions,*" Phys. Rev. B 92, 094410 (2015)

91 Zheng Zhu, Andrew J. Ochoa, Stefan Schnabel, Firas Hamze and **Helmut G. Katzgraber**, "*Best-case performance of quantum annealers on native spin-glass benchmarks: How chaos can affect success probabilities,*" Phys. Rev. A 93, 012317 (2016)

90 **Helmut G. Katzgraber**, Firas Hamze, Zheng Zhu, Andrew J. Ochoa, and H. Munoz-Bauza, "*Seeking Quantum Speedup Through Spin Glasses: The Good, the Bad, and the Ugly,*" Phys. Rev. X 5, 031026 (2015)

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